William Du

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EDUCATION

Carnegie Mellon University — Mountain View, United States
Master of Science in Software Engineering — December 2023

Queen's University — Kingston, Canada
Bachelor of Applied Science in Computer Engineering — May 2022
Major GPA: 3.9/4.0

PROFESSIONAL EXPERIENCE

Lucid Motors — Newark, United States
Software Engineer — Feb 2024 – Present

- Improved distributed automation framework performance by parallelizing execution across 45+ TCU devices, reducing full regression runtime by 85% and accelerating release validation cycles
- Developed and maintained 800+ automated tests using Python and Robot Framework, executed 1–3 times weekly per release, ensuring continuous validation of production software across NA, EU, and KSA
- Achieved an 80% reduction in system costs by creating a cost-effective design, reducing redundancies
- Designed and implemented long-duration stability and stress testing framework of CPU, memory, thermals, disk I/O, network, and service-level metrics, enabling proactive detection of performance degradation
- Managed large-scale Linux-based test farm infrastructure using Jenkins, GitLab CI, AWS, and MongoDB, supporting automated validation workflows across multiple engineering teams

Restaurant Brands International — Miami, United States
Software Engineer Intern — Jun 2023 – Aug 2023

- Deployed feature successfully to 52% of the total device population, benefiting 2.5M unique iPhone users
- Spearheaded the development of the iOS Live Activities feature and background fetching for Burger King
- Designed a custom package and patch to allow for proper linking and copying of files into native iOS app

PROJECTS

Machine Learning for Lithium-Ion Cathode Materials Discovery — Feb 2026 – Mar 2026

- Built a multi-sourced data pipeline aggregating 46K cathode materials data from Materials Project, OQMD, AFLOW, and JARVIS with automated de-duplication and leakage-safe composition splits
- Benchmarked 5 ML models (RF, XGBoost, CGCNN, M3GNet, TensorNet) for 4 battery properties (voltage, capacity, formation energy, stability), achieving $R^2 = 0.995$ on formation energy
- Engineered Magpie composition features (132-dim) and PBC-aware crystal graphs for GNN training (avg. 22 atoms/structure) using PyTorch Geometric with Gaussian distance edge features
- Implemented graph neural networks (CGCNN, M3GNet, TensorNet) using PyTorch Geometric and matgl, demonstrating performance gains from crystal-structure features over composition baselines
- Developed a Streamlit dashboard for cathode performance prediction with interactive 3D crystal visualization (py3Dmol) and automated lookup from a 46K-material database

Manga Buddy — Jan 2026 – Mar 2026

- Built a CV/NLP tool translating manga using a dual OCR pipeline (PaddleOCR + manga-ocr transformer)
- Implemented Union-Find clustering for text regions and collision-aware rendering for translated overlays
- Developed multi-threaded PyQt5 application with async ML inference to maintain a responsive UI

SKILLS

Python | Java | C/C++ | SQL | Linux | Scikit-learn | TensorFlow | OpenCV | React Native | TypeScript | Swift